

Probability Problems & Solutions

A worked collection of probability interview problems, with full explanations

18 probability problems, each with a step-by-step explanation of the reasoning behind the solution.

Problem 1 — Same Side Probability

Suppose we have two coins. One is fair and the other is biased where the probability of it coming up heads is $3/4$. Let's say we select a coin at random and flip it two times. What is the probability that both flips result in the same side?

Solution:

By the Law of Total Probability: $P(\text{Same}) = P(\text{Same} \mid \text{Biased}) \times P(\text{Biased}) + P(\text{Same} \mid \text{Fair}) \times P(\text{Fair})$

For the biased coin, probability that both flips result in the same side = $(3/4 \times 3/4) + (1/4 \times 1/4) = 9/16 + 1/16 = 10/16$.

For the fair coin, probability that both flips result in the same side = $(1/2 \times 1/2) + (1/2 \times 1/2) = 1/2$.

Final probability given a random selection of coins = $1/2 \times 10/16 + 1/2 \times 1/2 = 0.5625$.

Answer: 0.5625

Problem 2 — Estimated Rounds

Let's say there are six people trying to divide up into two equally separate teams. Because they want to pick random teams, on each round each person shows their hand in either a face-up or face-down position. If there are three of each position, then they'll split into teams. What's the expected number of rounds that everyone will have to pick a hand side before they split into teams?

Solution:

Since "they want to pick random teams" and there is no additional information given, we can assume that there is a 50/50 chance that each person puts a face down or face up. Thus the face of every individual person follows a Bernoulli distribution with probability of success $p = 0.5$. We are looking for the total number of face ups to be exactly 3, as that would imply that the rest of the group have their faces down. Let F denote the number of faces that are up. F is the sum of Bernoulli random variables and as such follows a binomial distribution, meaning the probability of $F = 3$ is:

$$P(F = 3) = C(6,3) (1/2)^3 (1/2)^3 = 0.3125$$

Let R be the number of rounds before having 3 faces up. Clearly, R follows a geometric distribution because it denotes the number of trials before one success. For geometric random variable G with success probability p , $E[G] = 1/p$. Thus, the expected number of rounds until teams form is:

$$E[R] = 1 / 0.3125 = 3.2$$

Answer: 3.2 rounds

Problem 3 — Ad Raters

Let's say we use people to rate ads. There are two types of raters. Random and independent from our point of view: 80% of raters are careful and they rate an ad as good (60% chance) or bad (40% chance). 20% of raters are lazy and they rate every ad as good (100% chance). Suppose we have 100 raters each rating one ad independently. What's the expected number of good ads?

Solution:

Let G be the event that the ad is rated good and C be the event a rater being careful.

$$P(G) = P(C \cap G) + P(C^c \cap G) = P(G | C) P(C) + P(G | C^c) P(C^c) = 0.8 \times 0.6 + 0.2 \times 1 = 0.68$$

The total number of good ads follows a binomial distribution because it asks for the result of many individual trials. In this case our parameters are number of trials $n = 100$ and success probability $p = 0.68$. The expected value of a binomial random variable is np , thus the expected number of ads rated good will be 68.

Sub-question 2: Now suppose we have 1 rater rating 100 ads. What's the expected number of good ads?

Let G_t be the total number of ads rated good.

$$\begin{aligned} E[G_t] &= P(C) E[G_t | C] + P(C^c) E[G_t | C^c] \\ &= 0.8 \times 100 \times 0.6 + 0.2 \times 100 \times 1 = 68 \end{aligned}$$

This is the same answer as question 1, showing that from a mathematical perspective, it doesn't matter if we have one random rater rating 100 ads or have 100 random raters rating one ad.

Sub-question 3: Suppose we have one ad, rated as bad. What's the probability that the rater was lazy?

$$\begin{aligned} P(C^c | G^c) &= [P(G^c | C^c) P(C^c)] / [P(G^c | C^c) P(C^c) + P(G^c | C) P(C)] \\ &= (0 \times 0.2) / (0 \times 0.2 + 0.8 \times 0.4) = 0 \end{aligned}$$

Answer: (a) 68 (b) 68 (c) 0

Problem 4 — First to Six

Amy and Brad take turns rolling a fair six-sided die. Whoever rolls a "6" first wins the game. Amy starts by rolling first. What's the probability that Amy wins?

Solution:

Probability that Amy wins = $P(\text{Amy rolls 6}) + P(\text{Amy doesn't roll 6}) \times P(\text{Brad doesn't roll 6}) \times P(\text{Amy rolls 6}) + \dots$

$$= 1/6 + (5/6 \times 5/6 \times 1/6) + (5/6 \times 5/6 \times 5/6 \times 5/6 \times 1/6) + \dots$$

$$= 1/6 + (5/6)^2 \times 1/6 + (5/6)^4 \times 1/6 + \dots$$

$$= (1/6) / (1 - (5/6)^2) = (1/6) / (1 - 25/36) = (1/6) / (11/36) = 1/6 \times 36/11 = 6/11$$

Answer: 6/11

Problem 5 — Jars and Coins

A jar holds 1000 coins. Out of all the coins, 999 are fair and one is double-sided with two heads. Picking a coin at random, you toss the coin ten times. Given that you see 10 heads, what is the probability that the coin is double headed and the probability that the next toss of the coin is also a head?

Solution:

Let F denote the event that the coin is fair and D denote the event that the coin is double headed.

$$P(\text{fair}) = 999/1000 = 0.999, P(\text{double-headed}) = 1/1000 = 0.001$$

Let H denote the event of getting a head.

$$P(D | 10H) = [P(10H | D) \cdot P(D)] / P(10H)$$

$$P(10H | D) = 1.$$

$$P(10H) = P(10H | D) P(D) + P(10H | F) P(F) = 1 \times 0.001 + (0.5)^{10} \times 0.999 = 0.00197558593$$

$$P(D | 10H) = (1 \times 0.001) / 0.00197558593 = 0.506$$

Probability next toss is a head:

$$P(\text{next toss of coin is a head}) = P(H | D) P(D) + P(H | F) P(F)$$

$$= P(\text{Heads} | \text{Double-headed coin}) P(\text{Double-headed coin}) + P(\text{Heads} | \text{fair coin}) P(\text{fair coin})$$

$$= 1 \times 0.506 + (1 - 0.506) \times 0.5 = 0.753$$

Here we are considering that the next toss of the coin is dependent on the previous 10 tosses.

Answer: $P(D | 10H) \approx 0.506$; $P(\text{next heads}) \approx 0.753$

Problem 6 — 500 Cards

Imagine a deck of 500 cards numbered from 1 to 500. If all the cards are shuffled randomly and you are asked to pick three cards, one at a time, what's the probability of each subsequent card being larger than the previous drawn card?

Solution:

Imagine this as a sample space problem ignoring all other distracting details. If you have to draw three different numbered cards without replacement, and they are all unique, then we are assuming that there will be effectively a lowest card, a middle card and a high card. Let's make it easy and assume that we drew the numbers 1, 2 and 3. In our scenario, if we drew (1, 2, 3), then that would be the winning scenario. But what's the full range of outcomes we could draw? Let's map out all the possibilities:

(3, 2, 1), (3, 1, 2), (2, 1, 3), (2, 3, 1), (1, 3, 2), (1, 2, 3).

So six possibilities in the total sample space. And only one of them is the partition that we want. Given this, the answer is 1/6.

The trick is not to be distracted by the size of the population. The population does not matter if you are looking at the order within the sample.

Answer: 1/6

Problem 7 — Second Ace

Let's say you have to draw two cards from a shuffled deck, one at a time. What is the probability that the second card is not an ace?

Solution:

Let A_1 = first card is an ace, A_2^c = second card is not an ace. By total probability:

$$P(A_2^c) = P(A_1) \times P(A_2^c | A_1) + P(A_1^c) \times P(A_2^c | A_1^c)$$

$$\text{Probability} = (4/52 \times 48/51) + (48/52 \times 47/51)$$

— first term: first card ace, second card not ace; second term: first card not ace, second card not ace —
 $= (48 \times 51) / (52 \times 51) = 48/52 \approx 0.923$

Answer: 48/52 $\approx$ 0.923

Problem 8 — Four Person Elevator

There are four people on the ground floor of a building that has five levels not including the ground floor. They all get into the same elevator. If each person is equally likely to get on any floor and they leave independently of each other, what is the probability that no two passengers will get off at the same floor?

Solution:

Probability that no two persons can get off on the same floor = Probability that each person has to get off on their own floor

= [Number of ways of assigning five floors to four different people without repetition] / [Number of ways of assigning five floors to four different people]

$$= (5 \times 4 \times 3 \times 2) / (5 \times 5 \times 5 \times 5) = 0.192$$

Answer: 0.192

Problem 9 — Dice Worth Rolling

Let's play a game. You are given two fair six-sided dice and asked to roll them. If the sum of the values on the dice equals seven, then you win \$21. However, you must pay \$10 for each roll. Is the game worth playing?

Solution:

Let D_1 and D_2 be the result of the first and second dice respectively. There are six ways that the sum ($D_1 + D_2$) = 7:

$D_1=1, D_2=6; D_1=2, D_2=5; D_1=3, D_2=4; D_1=4, D_2=3; D_1=5, D_2=2; D_1=6, D_2=1.$

$$P(D_1 + D_2 = 7) = 6/36 = 1/6.$$

We can use the expected value of $(D_1 + D_2)$ to get an estimate of the profit. Let's call the profit you make from this game P . The sample space of P is $\{11, -10\}$ because we either make \$21 by rolling a seven or lose \$10. Thus the possible results from the game are:

$$P(P = p) = \{ 11, \text{ with probability } 1/6 ; -10, \text{ with probability } 5/6 \}$$

$$\text{Expected value of } P \text{ is } E[P] = 11 \cdot (1/6) - 10 \cdot (5/6) \approx -6.5$$

Since $E[P] < 0$, we on average lose money from playing this game. Thus, this game is not worth playing.

Answer: $E[P] \approx -\$6.5$; not worth playing

Problem 10 — Biased five out of six

Let's say we are given a biased coin that comes up heads 30% of the time when tossed. What is the probability of the coin landing heads exactly 5 times out of 6 tosses?

Solution:

$$P(5 \text{ heads}) = C(6,5) (0.3)^5 (1 - 0.3)^{(6-5)} = 0.0102060$$

Answer: ≈ 0.0102

Problem 11 — Three Zebras

Three zebras are chilling in the desert. Suddenly a lion attacks. Each of the three zebras is sitting on a corner of an equilateral triangle. Each zebra randomly picks a direction and only runs along the outline of the triangle to either edge of the triangle. What is the probability that none of the zebras collide?

Solution:

The only possibility in which they fail to collide is if they choose to run in a clockwise direction or a counter-clockwise direction.

Probability that zebras will run in a clockwise direction = Probability that zebras will run in an anticlockwise direction = $1/2 \times 1/2 \times 1/2 = 1/8$.

Total probability = $1/8 + 1/8 = 1/4$.

Answer: $1/4$

Problem 12 — Raining in Seattle [CORRECTED]

You are about to get on a plane to Seattle. You want to know if you should bring an umbrella. You call 3 random friends of yours who live there and ask each independently if it's raining. Each of your friends has a $2/3$ chance of telling you the truth and a $1/3$ chance of messing with you by lying. All 3 friends tell you that "Yes" it is raining. What is the probability that it's actually raining in Seattle?

Solution:

We want $P(\text{rain} \mid \text{all three say "Yes"})$. This is a Bayes' rule problem and requires a prior probability of rain, which we call p . If it is raining, all three say "Yes" only when all tell the truth, so $P(3 \times \text{Yes} \mid \text{rain}) = (2/3)^3 = 8/27$. If it is not raining, all three say "Yes" only when all lie, so $P(3 \times \text{Yes} \mid \text{no rain}) = (1/3)^3 = 1/27$. By Bayes' rule:

$$P(\text{rain} \mid 3 \times \text{Yes}) = (8/27)p / [(8/27)p + (1/27)(1 - p)] = 8p / (7p + 1).$$

With an uninformed prior $p = 1/2$: $= 8(1/2) / (7(1/2) + 1) = 4 / 4.5 = 8/9 \approx 0.889$.

Answer: $8/9 \approx 0.889$

A common shortcut computes $1 - (1/3)^3 = 26/27$, but that is the probability that at least one friend is telling the truth — not the probability that it is raining. The correct approach uses Bayes' rule and requires a prior probability of rain; with an uninformed prior of $1/2$ the answer is $8/9$.

Problem 13 — Lazy Raters

Netflix has hired people to rate movies. Out of all the raters, 80% of the raters carefully rate movies and rate 60% of the movies as good and 40% as bad. The other 20% are lazy raters and rate 100% of the movies as good. Assuming all raters rate the same amount of movies, what is the probability that a movie is rated good?

Solution:

Let G be the event a movie is rated good. $P(G)$ is the probability of the event.

Prior probabilities:

- Given the rater is careful: 60% are rated good
- Given the rater is lazy: 100% are rated good
- $P(G \mid \text{careful rater}) = 0.6$
- $P(G \mid \text{lazy rater}) = 1$

Event probabilities:

- $P(\text{careful rater}) = 0.8$
- $P(\text{lazy rater}) = 0.2$

$$P(G) = P(G \mid \text{careful rater}) \times P(\text{careful rater}) + P(G \mid \text{lazy rater}) \times P(\text{lazy rater}) = 0.6 \times 0.8 + 1 \times 0.2 = 0.68$$

Answer: 0.68

Problem 14 — Secret Wins

There are 100 students that are playing a coin-tossing game. The students are given a coin to toss. If a student tosses the coin and it turns up heads, they win. If it comes up tails, they must flip again. If the coin comes up heads the second time, the students will lie and say they have won when they didn't. If it comes up tails, then they will say they have lost. If 30 students at the end say they won the game, how many students actually did win the game?

Solution:

Let W denote the number of students that actually won. The question lets us know that 70 students didn't say that they won. This means that $W \leq 30$. Consider the probability of someone winning given they said they won. Assuming the coin is fair, the probability of a student winning is 0.5. But the probability of a student saying they won is 0.75, since it includes the 0.25 (0.5×0.5) probability that a student flips heads on the second try and lies about winning. Thus:

$$P(\text{Won} \mid \text{Said won}) = P(\text{Won} \cap \text{Said won}) / P(\text{Said won}) = P(\text{Won}) / P(\text{Said won}) = 0.5 / (0.5 + 0.25) = 2/3$$

Since everyone that actually won said that they won, we can model the students who actually won as a binomial distribution with population size 30 and success probability $2/3$. Since if $X \sim B(n, p)$, then $E[X] = np$, our expected number of students who actually won is:

$$30 \times 2/3 = 20 \text{ students}$$

Answer: 20 students

Problem 15 — Marble Bucket [PART 2 CORRECTED]

We have two buckets full of marbles. There are 30 red marbles and 10 black marbles in Bucket 1 and 20 red and 20 black marbles in Bucket 2. Your friend secretly pulls a marble from one of the two buckets and shows you that the marble is red. What is the probability that it was pulled from Bucket 1?

Solution:

Bucket 1 = 10 Black + 30 Red

Bucket 2 = 20 Black + 20 Red

$$\begin{aligned} P(\text{Bucket} = 1 \mid \text{Marble} = \text{Red}) &= [P(\text{Marble} = \text{Red} \mid \text{Bucket} = 1) P(\text{Bucket} = 1)] / P(\text{Marble} = \text{Red}) \\ &= (3/4 \times 1/2) / (5/8) = (3/8) / (5/8) = 3/5 \end{aligned}$$

Sub-question 2: Let's say your friend puts the marble back in and now picks two marbles. She draws one marble, puts it back in the same bucket, and then draws a second. They both happen to be red. What is the probability that they both came from Bucket 1?

(Corrected solution:) Both draws come from the same unknown bucket, so we must condition on both reds jointly:

$$P(RR \mid B1) = (3/4)^2 = 9/16; P(RR \mid B2) = (1/2)^2 = 1/4$$

$$P(RR) = 1/2 \times 9/16 + 1/2 \times 1/4 = 9/32 + 4/32 = 13/32$$

$$P(B1 \mid RR) = (9/16 \times 1/2) / (13/32) = (9/32) / (13/32) = 9/13$$

Answer: (a) 3/5 (b) 9/13 $\approx$ 0.692

Because both marbles come from the same unknown bucket, the two reds must be conditioned on jointly. Treating them as independent posteriors — $(3/5)^2 = 9/25$ — is incorrect; the correct probability is 9/13.

Problem 16 — Coin Flip Probability

Let's say you are playing a coin flip game. You start with \$30. If you flip heads, you win one dollar, but if you get tails, you lose one dollar. You keep playing until you run out of money or you win \$100. What is the probability that you win \$100?

Solution:

This is a variation on an old applied probability problem called the Gambler's Ruin. We invoke a probabilistic structure called a random walk. The gambler can walk from \$30 to \$100 by taking numerous paths. Maybe he first walks from \$30 to \$29, or from \$30 to \$31. At each step in the walk, we can have two possible states we can go to next. Having k dollars, we can only walk to $(k+1)$ dollars or $(k-1)$ dollars in one step.

Our problem is a special case where moving to either of the possible next two states has equal probability of 0.5. Additionally, for each step, the previous outcome has no influence on the next. That is, each flip of the coin is an independent event. Because of these facts, we expect the gambler to have an equal number of wins and losses, thus remaining at \$30 in the long run. If we name our process X , then $E[X] = 30$.

Now, we know our walk has two possible end points and that we have 100% chance of reaching one or the other. The two end points are \$100 and \$0. Let's define the variable r as the probability of reaching \$100. Then our probability of reaching \$0 is $(1 - r)$.

The expected value of any discrete random variable or process Y is the sum of the products of the values of each event with the probability of their occurrence. That is:

$$E[Y] = \text{sum over } y \text{ of } [y \times P(y)].$$

In our case, we have $E[X] = 30$. Our two possible events are 100 and 0 with probability of occurrence r and $(1 - r)$ respectively.

$$E[X] = 100r + 0(1 - r) \rightarrow 30 = 100r \rightarrow r = 0.3.$$

So probability of winning is 30%.

Answer: 30%

Problem 17 — Fake Algorithm Reviews

Let's say we are trying to determine fake reviews on our products. Based on past data, 98% reviews are legitimate and 2% are fake. If a review is fake, there is a 95% chance that the machine learning algorithm identifies it as fake. If a review is legitimate, there is a 90% chance that the machine learning algorithm identifies it as legitimate. What is the percentage chance that the review is actually fake when the algorithm detects it as fake?

Solution:

Let L be the event that a review is actually legitimate and I be the event that the review is identified as legitimate.

$$P(L) = 0.98, P(I^c | L^c) = 0.95, P(I | L) = 0.9$$

By Bayes' Rule:

$$\begin{aligned} P(L^c | I^c) &= [P(I^c | L^c) P(L^c)] / [P(I^c | L^c) P(L^c) + P(I^c | L) P(L)] \\ &= [P(I^c | L^c) P(L^c)] / [P(I^c | L^c) P(L^c) + (1 - P(I | L)) P(L)] \\ &= (0.95 \times 0.02) / (0.95 \times 0.02 + (1 - 0.9) \times 0.98) \\ &= 0.019 / (0.019 + 0.098) = 0.019 / 0.117 \approx 0.162 \end{aligned}$$

Answer: $\approx 16.2\%$

Problem 18 — Probability of picking a biased coin

Suppose you have a bag of 100 coins of which 1 is biased with both sides as heads. You pick a coin from the bag and toss it three times. The result of all three tosses is heads. What is the probability that the selected coin is biased?

Solution:

Let A be the event that the selected coin is biased and B be the event that the result of three tosses is heads.

$$P(A | B) = [P(B | A) P(A)] / [P(B | A) P(A) + P(B | A^c) P(A^c)]$$

$$P(B | A) = 1.$$

$$P(A) = 1/100.$$

$$P(A^c) = 1 - 1/100 = 99/100.$$

$$P(B | A^c) = (1/2)^3 = 1/8.$$

$$P(A | B) = [1 \times 1/100] / [1 \times 1/100 + 1/8 \times 99/100] = (1/100) / [(1/100)(1 + 99/8)] = 8/107$$

Answer: $8/107 \approx 0.075$